

What to Do

Model Problem Solving — Suggested time 10 minutes

1. Tell your class that they are going to be working today to design a spacecraft to go to Saturn.
2. Read “Summary of Spacecraft Design Problems” aloud to your class.
3. Ask students to turn to a partner and spend a few minutes brainstorming a way to keep the spacecraft from burning up when it leaves Earth.
4. Collect ideas from the class.
5. Tell your students that they will be put into pairs to work as a design team and will be given a worksheet to complete, and that at the end of the class, each team will choose one or two things from their worksheet to share with the other students.
6. Give each pair of students a copy of the “Designing Your Spacecraft Worksheet.”

Design the Spacecraft — Suggested time 35 minutes

1. Ask the design teams to proceed in designing their spacecraft.
2. Circulate throughout the room to assist the students as they complete their worksheets.

Share with the Class — Suggested time 15 minutes

1. Have students choose a part of their spacecraft designs to share with their classmates, and decide how they will present together (who shares what part).
2. Ask the design team pairs to come up and present something from their worksheets.
3. Have the class share things they like about each other’s work, questions they have, and suggestions for the presented designs.
4. Ask students to put their name and date on their work, decide which partner will store it, and put it inside the Saturn Discovery Log. (You may want to offer to make a copy of the work, so that each student can have the work in his/her Saturn Discovery Log.)

Extensions

Write a Descriptive Paragraph

1. Have the students write a paragraph describing everything about their spacecraft.
2. For an extra challenge, you can hang the sketches of spacecraft around the room, and have the class work in pairs to match descriptive paragraphs to the actual sketches.

